

controllers to distinguish between enemy aircraft and their own. Controllers adapted the techniques developed during World War I – keeping track of aircraft movements by pushing models around on a chart and using two-way radio links to give instructions to pilots in the air – to direct interceptors against intruding bombers.

Britain was especially well placed to develop a radar-based defense system because its front line was the coast, and radar worked much better over the uncluttered sea than over land. The British

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HAROLD MACMILLAN

FORMER BRITISH PRIME MINISTER

were also highly motivated by fear of air attack. The age of the bomber had come as a far greater psychological shock to Britain than to any other country because, protected by the Royal Navy, its people had long thought themselves immune to attack from abroad.

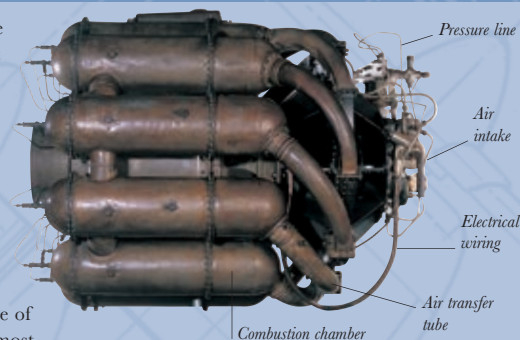
The British government was told by its aviation chiefs that it could expect 20,000 civilian casualties in London on the first day of a war with Germany, and 150,000 in the first week (in fact, there were 295,000 casualties from air attacks in the whole of Britain in six years of war

THE FIRST JETS

ON AUGUST 27, 1939, four days before the start of World War II, the first jet aircraft took off from Marienehe airfield in Germany. Test pilot Erich Warsitz kept the diminutive Heinkel He 178 in the air for just six minutes, but it was enough to open a new era in aviation. Aircraft manufacturer Ernst Heinkel commented: “The hideous wail of the engine was music to our ears.”

The wailing turbojet engine was the brainchild of Hans von Ohain, a graduate of the University of Göttingen, Germany’s most prestigious center of theoretical aeronautics. It used a gas turbine to generate thrust in accordance with Newton’s Third Law of Motion. The aircraft scooped up air as it moved along; this air was compressed, combined with fuel, and ignited; and the jet of hot gas forced out of the back of the engine propelled the aircraft forward.

Ohain was only one of a number of researchers in the 1930s investigating the use of gas turbines to create jet propulsion. Another was Flight Lieutenant Frank Whittle of the RAF, who filed his first patent for a



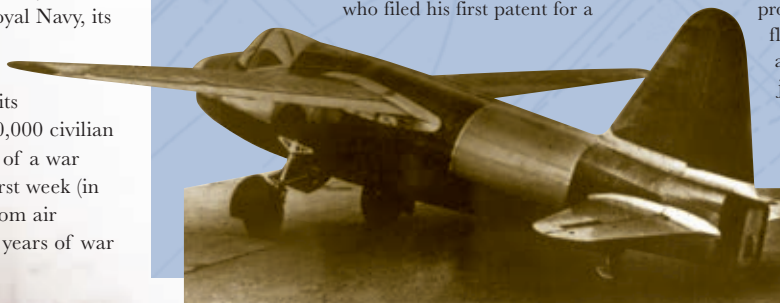
JET PROPULSION ENGINE

Whittle’s jet engine worked by sucking in air, then compressing and burning it with fuel, to create thrust. Faster than the propeller engine, it was also more economical on fuel.

jet engine as early as 1930. But whereas Whittle struggled to develop his engine in the face of financial difficulties and official indifference, Ohain and his assistant, Max Hahn, were backed both by Heinkel and the German Air Ministry. As a result, by the time Britain’s first jet prototype, the Gloster E.28/39, flew in 1941, Germany was already developing practical jet-propelled warplanes.

FIRST JET AIRCRAFT

In August 1939 Erich Warsitz flew the world’s first jet aircraft, an He 178, reaching a top speed of over 400mph (640kph).



from 1939 to 1945). Not surprisingly, faced with such alarmist predictions, the defenders of democracy in Europe were unnerved by fears of air attack as Hitler pressed for advantage in 1938. The last thing they wanted was a war that would begin with the immediate destruction of cities and hundreds of thousands of civilian casualties.

Looking back on that time, British politician Harold Macmillan commented: “We thought of air warfare in 1938 rather as people think of nuclear warfare today.” In other words, it was expected to lead swiftly not only to mass slaughter but also to a total breakdown of civilization. With war on the horizon, plans were finalized in 1939 for the distribution of gas masks, provision of bomb shelters, and evacuation of children. In the run-up to the war, filmgoers who saw the popular science-fiction movie *Things to Come* (1936), based on a 1933 novel by H.G. Wells, found in its image of a world laid

to waste by aerial bombardment a very plausible version of their own future. Watching warplanes maneuvering over rural England in 1932, British poet Siegfried Sassoon had predicted that, one day, “fear will be synonymous with flight.” For many, that day had come.

AERIAL APOCALYPSE

The film version of H.G. Wells’ novel *The Shape of Things to Come* vividly depicts the earth laid to waste by aerial bombardment in a 20-year war.

